Identifying Cell Type-Specific Chemokine Correlates with Hierarchical Signal Extraction from Single-Cell Transcriptomes

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Introduction

Biological data is inherently heterogeneous and high-dimensional. Single-cell RNA-sequencing (scRNA-seq) of transcripts in a tissue sample generates data for thousands of cells, each of which is characterized by upwards of tens of thousands of genes. How to identify the subsets of cells and genes that are associated with a label of interest remains an open question.

In this work, we integrate a signal-extractive neural network architecture with axiomatic feature attribution to classify tissue samples based on single-cell gene expression profiles.

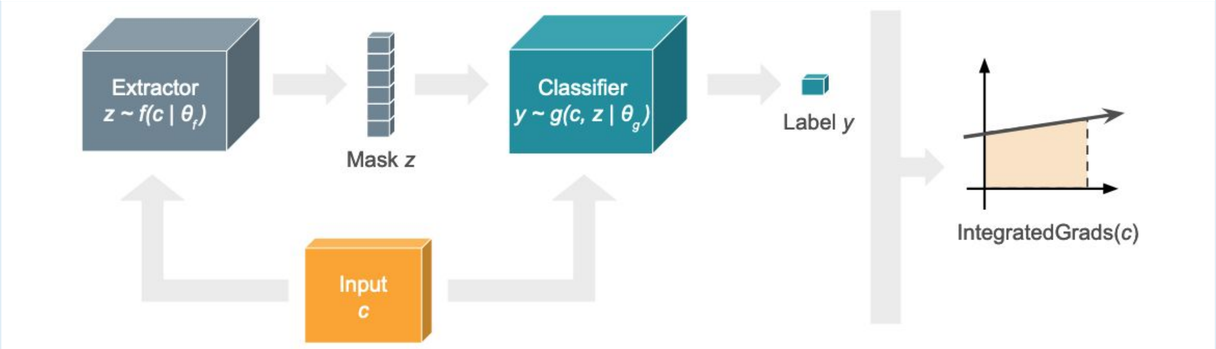
We demonstrate applicability of this approach in one synthetic setting and two real-world settings for discovering cell type-specific chemokine correlates: predicting response to PD-1 immune checkpoint inhibitor in multiple tissue types and classifying DNA mismatch repair status in colorectal cancer. Our approach not only significantly outperforms traditional machine learning classifiers but also presents actionable biological hypotheses of chemokine-mediated tumor immunogenicity.

Background

Chemokines are chemotactic cytokines that mediate cell migration and positioning in both tissue and lymph nodes, particularly with respect to immune cells. Once a chemokine receptor is activated by its cognate chemokine ligand, the chemokine receptor-expressing cell migrates up the associated chemokine concentration gradient towards the source of chemokine production.

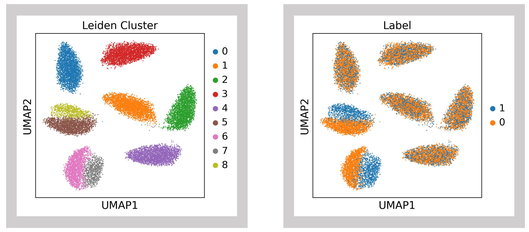
In the context of cancer, chemokines can promote pro-tumorigenic activity, such as tumor proliferation, anti-apoptosis, and metastasis. Furthermore, chemokines can mediate immune evasion by influencing T cell sequestration in stroma or by cultivating a microenvironment of high-density non-activating immune cells, leading to inefficient T cell search of malignant cells. The importance of chemokines to tumorigenicity warrants further study of these molecules, their role in the tumor-immune microenvironment, and their role in response to immunotherapy.

Method



Application 1. Binary classification with *in silico* scRNA-seq data

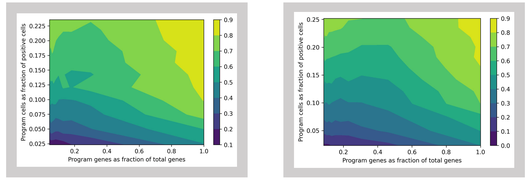
Using a Gamma-Poisson generative model, we simulate scRNA-seq data, where positive-labeled tissue samples contain cells from clusters 7 and 8 and negative-labeled tissue samples do not.



SCIONN outperforms standard machine learning models on not only classification performance but also signal extraction accuracy.

Gene Set	Model	Loss	AUC	Number of Correct Genes as Fraction of Top 120 Genes	Genes $> \mu + 2\sigma$
Same	LogReg	0.2470 ± 0.1415	0.9813 ± 0.0720	0.8137 ± 0.1251	0.9257 ± 0.1523
	RNN	0.1331 ± 0.1891	0.9863 ± 0.0399	0.2210 ± 0.0700	0.2518 ± 0.0850
	CNN	0.1516 ± 0.0132	1.0000 ± 0.0000	0.5820 ± 0.0666	0.6302 ± 0.0876
	SCIONN	$\mathbf{0.1858 \pm 0.0325}$ *	$\mathbf{1.0000 \pm 0.0000}$ * +	$\mathbf{0.8160 \pm 0.0215}$ + ⋆	$\mathbf{0.9067 \pm 0.0303}$ + ⋆
Different	LogReg	0.3812 ± 0.1560	0.9436 ± 0.1272	0.6812 ± 0.1477	0.8121 ± 0.1965
	RNN	0.2822 ± 0.3654	0.9539 ± 0.0862	0.1983 ± 0.0590	0.2193 ± 0.0687
	CNN	0.1710 ± 0.0209	1.0000 ± 0.0000	0.6103 ± 0.0407	0.7065 ± 0.0871
	SCIONN	$\mathbf{0.2261 \pm 0.0382}$ *	$\mathbf{1.0000 \pm 0.0000}$ * +	$\mathbf{0.7740 \pm 0.0321}$ * + ⋆	$\mathbf{0.8976 \pm 0.0340}$ * + ⋆

SCIONN is robust to noise, requiring just 5% of genes and 23% of cells in an *in silico* tissue sample to encode signal in order to distinguish signal from noise with greater than 70% accuracy.



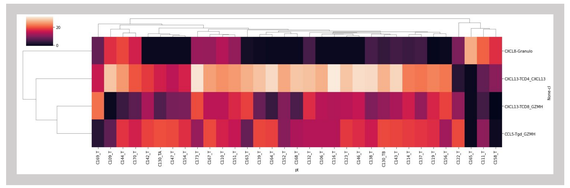
Application 2. DNA mismatch repair deficiency classification

DNA mismatch repair deficiency is an important phenotype for determining treatment decisions for cancer patients. SCIONN outperforms standard machine learning classifiers on DNA mismatch repair status classification, independently validated on multiple public datasets.

Model	Loss	AUC
LogReg	0.6849 ± 0.1173	0.6040 ± 0.2477
RNN	0.6878 ± 0.0204	0.5408 ± 0.0782
LSTM	0.6250 ± 0.0795	0.7573 ± 0.1292
GRU	0.6406 ± 0.0732	0.7023 ± 0.1258
SCIONN	$\mathbf{0.5320 \pm 0.0679}$ * ◇ ⋆○	$\mathbf{0.8701 \pm 0.0954}$ * ◇ ⋆○

Independent Dataset	Model	Loss	AUC
GSE146771_colon10x	LogReg	1.3730 ± 0.6203	0.4934 ± 0.1323
	RNN	0.7004 ± 0.0555	0.4993 ± 0.1495
	LSTM	0.6425 ± 0.1082	0.6194 ± 0.2475
	GRU	0.5878 ± 0.1222	0.6419 ± 0.1914
	SCIONN	$\mathbf{0.3920 \pm 0.1021}$ * ◇ ⋆○	$\mathbf{0.8691 \pm 0.2201}$ * ◇ ⋆○
GSE146771_colonSS2	LogReg	0.7878 ± 0.1469	0.5328 ± 0.0842
	RNN	0.6967 ± 0.0313	0.5302 ± 0.0859
	LSTM	0.6623 ± 0.0811	0.7197 ± 0.1864
	GRU	0.6431 ± 0.1001	0.6848 ± 0.1603
	SCIONN	$\mathbf{0.5030 \pm 0.0897}$ * ◇ ⋆○	$\mathbf{0.9041 \pm 0.0843}$ * ◇ ⋆○
GSE132465_colon10x	LogReg	0.8509 ± 0.1503	0.5469 ± 0.0541
	RNN	0.6974 ± 0.0346	0.5074 ± 0.0676
	LSTM	0.7574 ± 0.1474	0.6333 ± 0.0821
	GRU	0.6915 ± 0.0823	0.5950 ± 0.0964
	SCIONN	$\mathbf{0.6711 \pm 0.0960}$ * ◇ ⋆	$\mathbf{0.7275 \pm 0.0321}$ * ◇ ⋆○

Expression of the chemokine CXCL13 in activated CD4+ and CD8+ T cells was assigned the highest attribution for DNA mismatch repair deficiency, which persisted in independent datasets and aligns with published findings.

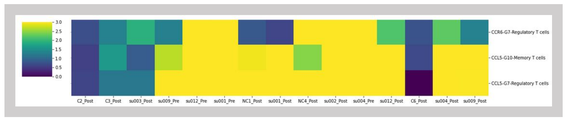


Application 3. PD-1 checkpoint inhibitor response prediction

PD-1 checkpoint inhibitor is one of the most promising cancer treatments to date, offering durable response for a subset of patients. However, predicting which patients will respond to treatment, and why, is not well understood. SCIONN outperforms standard machine learning classifiers on PD-1 inhibitor response prediction.

Model	Loss	AUC
LogReg	0.6322 ± 0.0652	0.7119 ± 0.1116
RNN	0.6484 ± 0.0643	0.6443 ± 0.1340
LSTM	0.6362 ± 0.1063	0.6889 ± 0.1662
GRU	0.6292 ± 0.0900	0.7028 ± 0.1387
SCIONN	$\mathbf{0.6144 \pm 0.0884}$ * ◇ ⋆○	$\mathbf{0.7583 \pm 0.1588}$ * ◇ ⋆○

Based on the attribution scores, expression of the chemokine CCL5 by memory T cells and regulatory T cells consistently scored high on attribution for PD-1 inhibitor responders, consistent with recent *in vivo* model findings.



Conclusion

We introduce a framework for identifying cell type-specific chemokine correlates of response to immune checkpoint inhibitors in multiple tissues and DNA mismatch repair status in colorectal cancers. By employing a deep neural network that couples signal extraction with label classification, we successfully identify signals derived from cell type-gene tuples that would otherwise have been overlooked under poorer-performing neural network architectures. We foresee this framework extending to other classification problems where the correlation structure is learned from hierarchically-structured data, such as spatially-resolved single-cell transcriptomics.

Acknowledgements

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